# Econometrics

## Ridge and Lasso Regression

Jose Angel Garcia Sanchez

Université Paris 1 Panthéon-Sorbonne



UNIVERSITÉ PARIS 1

PANTHÉON SORBONNE

# Outline


1. Introduction to Regularization
2. Ridge Regression
3. Lasso Regression
4. Elastic Net
5. Choosing the Tuning Parameter
6. Inference and Extensions
7. Practical Implementation

# Motivation

- OLS minimizes the sum of squared residuals: $\min_\beta \sum_{i=1}^{n}(y_i - x_i'\beta)^2$
- OLS works well when $n \gg p$ (many observations, few variables)
- But OLS can fail when:
  - $p$ is large relative to $n$ (high-dimensional data); if $p > n$, $X'X$ is not even invertible
  - There is multicollinearity among regressors
  - We want to improve out-of-sample prediction
- Solution: add a *penalty term* to the objective function
- This is called *regularization* or *shrinkage*

# The Bias-Variance Trade-off

- Recall that the Mean Squared Error of an estimator can be decomposed: $MSE = Bias^2 + Variance$
- OLS is unbiased but can have high variance, especially when:
  - Predictors are highly correlated (multicollinearity)
  - The number of predictors is large
- Regularization introduces some bias but reduces variance
- If the reduction in variance exceeds the increase in squared bias, we get a lower MSE
- This improves prediction accuracy, especially out-of-sample

# The General Penalized Regression Framework

- General form: $\min_\beta \left\{ \sum_{i=1}^n (y_i - x_i'\beta)^2 + \lambda \cdot P(\beta) \right\}$
- $\lambda \geq 0$ is the *tuning parameter* (or regularization parameter)
- $P(\beta)$ is the *penalty function*
- When $\lambda = 0$: we get OLS
- When $\lambda \to \infty$: all coefficients shrink to zero
- Different choices of $P(\beta)$ give different estimators:
  - Ridge: $P(\beta) = \sum_{j=1}^p \beta_j^2$ (L2 penalty)
  - Lasso: $P(\beta) = \sum_{j=1}^p |\beta_j|$ (L1 penalty)
  - Elastic Net: combination of both

# Important Preliminary: Standardization

- Regularization penalizes the size of coefficients
- But coefficient size depends on the scale of variables
- A variable measured in millions will have a tiny coefficient compared to one measured in units
- Solution: *standardize* all variables before estimation
- For each variable $x_j$: $\tilde{x}_j = \frac{x_j - \bar{x}_j}{s_{x_j}}$
- After standardization, all variables have mean 0 and standard deviation 1
- The intercept is typically not penalized
- After estimation, coefficients can be transformed back to the original scale if needed

# Ridge Regression: Definition

- Ridge regression adds an L2 penalty to the OLS objective:

$$\hat{\beta}^{Ridge} = \arg\min_{\beta} \left\{ \sum_{i=1}^{n} (y_i - x_i'\beta)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \right\}$$

- Equivalently, in matrix notation:

$$\hat{\beta}^{Ridge} = \arg\min_{\beta} \left\{ (Y - X\beta)'(Y - X\beta) + \lambda \beta'\beta \right\}$$

- The penalty term $\lambda \sum_{j=1}^{p} \beta_j^2$ shrinks coefficients toward zero
- Large coefficients are penalized more heavily (quadratic penalty)

UNIVERSITÉ PARIS 1
PANTHÉON SORBONNE

# Ridge Regression: Closed-Form Solution

- Taking the first-order condition:

$$\frac{\partial}{\partial \beta}\left[(Y - X\beta)'(Y - X\beta) + \lambda\beta'\beta\right] = 0$$

- We get: $-2X'Y + 2X'X\beta + 2\lambda\beta = 0$
- Solving for $\beta$:

$$\hat{\beta}^{Ridge} = (X'X + \lambda I)^{-1}X'Y$$

- Compare with OLS: $\hat{\beta}^{OLS} = (X'X)^{-1}X'Y$
- The term $\lambda I$ is added to $X'X$ before inversion
- This ensures $(X'X + \lambda I)$ is always invertible for $\lambda > 0$, even when $X'X$ is singular

# Ridge Regression: Properties (1)

- The Ridge estimator is biased: $E(\hat{\beta}^{Ridge}) \neq \beta$
- The bias is: $Bias(\hat{\beta}^{Ridge}) = -\lambda(X'X + \lambda I)^{-1}\beta$
- The variance is: $Var(\hat{\beta}^{Ridge}) = \sigma^2(X'X + \lambda I)^{-1}X'X(X'X + \lambda I)^{-1}$
- As $\lambda$ increases:
  - Bias increases (coefficients shrink more toward zero)
  - Variance decreases (estimates become more stable)
- There exists an optimal $\lambda^*$ that minimizes MSE, and it is always possible to beat OLS: some $\lambda > 0$ gives a lower MSE than OLS (Hoerl and Kennard, 1970)

# Ridge Regression: Properties (2)

- Ridge does NOT set coefficients exactly to zero
- All predictors remain in the model (no variable selection)
- Ridge is particularly useful when:
  - Predictors are highly correlated (multicollinearity)
  - We believe all predictors are relevant
  - We want to stabilize estimates
- The ridge penalty can be viewed as a Bayesian prior: $\beta \sim N(0, (\sigma^2/\lambda)I)$

# Ridge Regression: Geometric Interpretation

- Ridge regression can be written as a constrained optimization:

$$\min_{\beta} \sum_{i=1}^{n} (y_i - x_i'\beta)^2 \quad \text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 \leq t$$

- The constraint region is a ball (a disk in 2 dimensions)
- The OLS solution is at the center of the elliptical contours of the RSS
- The ridge solution is where the elliptical contours first touch the ball
- Because the ball has no corners, the solution typically has all coefficients non-zero
- Smaller $t$ (equivalently, larger $\lambda$) means more shrinkage

# Ridge and Multicollinearity

- Recall: with perfect multicollinearity, $X'X$ is singular and OLS fails
- With near-perfect multicollinearity, $(X'X)^{-1}$ has very large elements
- This leads to inflated variances and unstable OLS estimates
- Ridge regression solves this by adding $\lambda I$ to $X'X$
- Even if $X'X$ is singular, $(X'X + \lambda I)$ is always invertible for $\lambda > 0$
- The eigenvalues of $(X'X + \lambda I)$ are $d_j + \lambda$, where $d_j$ are the eigenvalues of $X'X$
- Small eigenvalues (source of instability) become at least $\lambda$ instead of nearly zero

# Lasso Regression: Definition

- Lasso = Least Absolute Shrinkage and Selection Operator (Tibshirani, 1996)
- Lasso adds an L1 penalty to the OLS objective:

$$\hat{\beta}^{Lasso} = \arg\min_{\beta} \left\{ \sum_{i=1}^{n} (y_i - x_i'\beta)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\}$$

- The key difference from Ridge: absolute values instead of squares
- This seemingly small change has profound implications
- Lasso performs both shrinkage AND variable selection

# Lasso: No Closed-Form Solution

- Unlike Ridge, Lasso has no closed-form solution
- The absolute value function $|\beta_j|$ is not differentiable at $\beta_j = 0$
- Optimization requires iterative algorithms:
  - Coordinate descent (most common)
  - LARS (Least Angle Regression)
  - Proximal gradient methods
- With an orthonormal design ($X'X = I$), each coefficient is a *soft-thresholding* of the OLS one:

$$\hat{\beta}_j^{Lasso} = \text{sign}(\hat{\beta}_j^{OLS}) \left( |\hat{\beta}_j^{OLS}| - \lambda/2 \right)_+$$

- Where $(x)_+ = \max(0, x)$: small OLS coefficients are set exactly to zero
- Compare Ridge in the same case: $\hat{\beta}_j^{Ridge} = \hat{\beta}_j^{OLS}/(1 + \lambda)$, a proportional shrinkage that never reaches zero

# Lasso: Variable Selection Property

- The key feature of Lasso: it sets some coefficients exactly to zero
- This performs automatic variable selection
- As $\lambda$ increases, more coefficients become exactly zero
- The model becomes more sparse (fewer non-zero coefficients)
- This is very useful when:
  - We have many potential predictors
  - We believe only a subset are truly relevant
  - We want an interpretable, parsimonious model
- Lasso identifies which variables matter and estimates their effects simultaneously

# Lasso: Geometric Interpretation

- Lasso can be written as a constrained optimization:

$$\min_{\beta} \sum_{i=1}^{n} (y_i - x_i'\beta)^2 \quad \text{subject to} \quad \sum_{j=1}^{p} |\beta_j| \leq t$$

- The constraint region is a diamond (or cross-polytope in higher dimensions)
- The diamond has corners on the axes
- The elliptical RSS contours often touch the diamond at a corner
- At a corner, one or more coefficients are exactly zero
- This is why Lasso produces sparse solutions

# Lasso: Properties

- The Lasso estimator is biased (like Ridge)
- Lasso is consistent for variable selection under certain conditions
- However, Lasso has some limitations:
  - With highly correlated predictors, Lasso tends to select one and ignore the others
  - When $p > n$, Lasso selects at most $n$ variables
  - Lasso estimates for selected variables are biased toward zero
- These limitations motivate extensions like Elastic Net and Adaptive Lasso

# Comparing Ridge and Lasso

| Property | Ridge | Lasso |
|---|---|---|
| Penalty | $\sum \beta_j^2$ (L2) | $\sum \lvert\beta_j\rvert$ (L1) |
| Closed-form solution | Yes | No |
| Variable selection | No | Yes |
| Handles multicollinearity | Yes | Partially |
| Correlated predictors | Shrinks together | Selects one |
| When $p > n$ | Works | Selects $\leq n$ |
| Bayesian interpretation | Normal prior | Laplace prior |

# Elastic Net: Combining Ridge and Lasso

- Elastic Net (Zou and Hastie, 2005) combines L1 and L2 penalties:

$$\hat{\beta}^{EN} = \arg\min_{\beta} \left\{ \sum_{i=1}^{n} (y_i - x_i'\beta)^2 + \lambda_1 \sum_{j=1}^{p} |\beta_j| + \lambda_2 \sum_{j=1}^{p} \beta_j^2 \right\}$$

- Alternative parameterization with mixing parameter $\alpha \in [0, 1]$:

$$\hat{\beta}^{EN} = \arg\min_{\beta} \left\{ \sum_{i=1}^{n} (y_i - x_i'\beta)^2 + \lambda \left[ \alpha \sum_{j=1}^{p} |\beta_j| + (1-\alpha) \sum_{j=1}^{p} \beta_j^2 \right] \right\}$$

- $\alpha = 1$: Lasso; $\alpha = 0$: Ridge

# Elastic Net: Properties

- Combines the best of both worlds:
  - Variable selection (from Lasso)
  - Grouping effect (from Ridge)
- With correlated predictors, Elastic Net tends to select groups of correlated variables together
- Can select more than $n$ variables when $p > n$
- Particularly useful when:
  - Predictors are highly correlated
  - We want both sparsity and stability
  - The number of predictors exceeds the sample size

# The Role of $\lambda$

- The tuning parameter $\lambda$ controls the amount of regularization
- $\lambda = 0$: no penalty, we get OLS
- $\lambda \to \infty$: maximum penalty, all coefficients go to zero
- Intermediate $\lambda$: trade-off between bias and variance
- How to choose $\lambda$? Several approaches:
  - Cross-validation (most common)
  - Information criteria (AIC, BIC)
  - Theoretical considerations

# Cross-Validation (1)

- K-fold cross-validation is the standard approach
- Procedure:
  1. Divide data into $K$ roughly equal parts (folds)
  2. For each fold $k = 1, \ldots, K$:
     - Fit the model on all data except fold $k$
     - Compute the prediction error on fold $k$
  3. Average the $K$ prediction errors
- Repeat for a grid of $\lambda$ values
- Choose the $\lambda$ that minimizes the cross-validation error
- Common choices: $K = 5$ or $K = 10$

# Cross-Validation (2)

- The CV error as a function of $\lambda$ typically has a U-shape
- Small $\lambda$: high variance, low bias (overfitting)
- Large $\lambda$: low variance, high bias (underfitting)
- Two common choices:
  - $\lambda_{min}$: minimizes the CV error
  - $\lambda_{1se}$: largest $\lambda$ whose CV error is within 1 standard error of the minimum
- $\lambda_{1se}$ gives a more parsimonious model (more regularization)
- This is the "one-standard-error rule"

# Information Criteria

- Alternative to cross-validation: use information criteria
- For Lasso, we can use:
  - AIC: $AIC = n \log(RSS/n) + 2\, df$
  - BIC: $BIC = n \log(RSS/n) + \log(n) \cdot df$
- The degrees of freedom of the Lasso can be estimated by the number of non-zero coefficients
- For Ridge: $df = \sum_{j=1}^{p} \frac{d_j}{d_j + \lambda}$, where $d_j$ are the eigenvalues of $X'X$
- BIC tends to select sparser models than AIC
- Cross-validation is generally preferred in practice

# Inference After Selection

- Standard inference (t-tests, confidence intervals) is invalid after Lasso selection
- Why? The selection process introduces additional randomness
- Naive standard errors are too small (underestimate uncertainty)
- Confidence intervals have incorrect coverage
- Several approaches have been developed:
  - Sample splitting
  - Selective inference
  - Debiased/Desparsified Lasso
  - Bootstrap methods
  - Post-double-selection (Belloni, Chernozhukov and Hansen, 2014), for the effect of one variable of interest in the presence of many controls

# Sample Splitting

- Simple approach to valid inference:
  1. Split the sample randomly into two parts
  2. Use the first part for variable selection (Lasso)
  3. Use the second part for estimation and inference (OLS on selected variables)
- Advantages: simple, valid inference
- Disadvantages:
  - Loses statistical power (uses only half the data for each step)
  - Results depend on the random split
- Can average over multiple random splits

# Post-Lasso OLS

- Also called "Post-Selection OLS" or "OLS post Lasso"
- Procedure:
  1. Run Lasso to select variables (non-zero coefficients)
  2. Run OLS using only the selected variables
- The OLS estimates are less biased than Lasso estimates
- Removes the shrinkage bias for selected variables
- Caution: standard errors from this OLS are still not valid for inference
- Valid inference requires additional corrections

# Adaptive Lasso

- Standard Lasso penalizes all coefficients equally
- This can lead to too much bias for large coefficients
- Adaptive Lasso (Zou, 2006) uses weighted penalties:

$$\hat{\beta}^{ALasso} = \arg\min_{\beta} \left\{ \sum_{i=1}^{n} (y_i - x_i'\beta)^2 + \lambda \sum_{j=1}^{p} w_j |\beta_j| \right\}$$

- Weights: $w_j = 1/|\hat{\beta}_j^{init}|^\gamma$ for some initial estimator $\hat{\beta}^{init}$ (e.g. OLS or Ridge)
- Typically $\gamma = 1$ or $\gamma = 2$
- Variables with small initial estimates are penalized more
- Under some conditions, Adaptive Lasso has the "oracle property": asymptotically, it selects the correct model and estimates its coefficients as if the true model were known

# Implementation in Software

- Ridge and Lasso are implemented in all major statistical software:
  - R: `glmnet` package (gold standard)
  - Python: `scikit-learn` (`Ridge`, `Lasso`, `ElasticNet`, and `RidgeCV`, `LassoCV`, `ElasticNetCV` with built-in cross-validation)
  - Stata: `lasso`, `elasticnet` commands
- `glmnet` uses coordinate descent, computes the entire solution path (all values of $\lambda$) efficiently, and standardizes the variables by default
- scikit-learn does **not** standardize: use `make_pipeline(StandardScaler(), LassoCV())`
- scikit-learn calls $\lambda$ `alpha`, and divides the Lasso and Elastic Net RSS by $2n$: its `alpha` is $\lambda/(2n)$ in our notation (but $\lambda$ for `Ridge`); our mixing parameter $\alpha$ is its `l1_ratio`

# Practical Recommendations

- Always standardize predictors before regularization
- Use cross-validation to choose $\lambda$
- Consider the one-standard-error rule for more parsimony
- If predictors are highly correlated, consider Elastic Net
- For inference, use appropriate methods (not naive standard errors)
- Check model performance on held-out test data
- Compare with OLS on a subset of variables (sanity check)
- Remember: regularization is primarily for prediction, not causal inference (unless using methods designed for it, such as post-double-selection)

# When to Use What?

- **OLS**: $n \gg p$, no multicollinearity, focus on inference
- **Ridge**: multicollinearity, all predictors believed relevant, prediction focus
- **Lasso**: many predictors, only some believed relevant, want variable selection
- **Elastic Net**: many correlated predictors, want both selection and grouping
- **Adaptive Lasso**: want consistent variable selection with oracle properties
- In practice: try several methods and compare via cross-validation

# Summary

- Regularization adds a penalty to the OLS objective function
- Ridge (L2): shrinks coefficients, handles multicollinearity, no selection
- Lasso (L1): shrinks and selects variables, produces sparse models
- Elastic Net: combines Ridge and Lasso benefits
- The tuning parameter $\lambda$ is chosen via cross-validation
- Standard inference is invalid after Lasso selection
- These methods are powerful tools for prediction in high-dimensional settings
- But remember: regularization introduces bias, which may not be desirable for causal inference